

STA 332/MATH 343: Homework 3 – Solutions

Duke University, Fall 2026

Due: September 25th, 11:59 PM on Gradescope

Constructing Estimators

1. Let $X_1, \dots, X_n \stackrel{\text{i.i.d.}}{\sim} \text{Gamma}(\alpha, \beta)$, where both the shape $\alpha > 0$ and the rate $\beta > 0$ are unknown. Find the method-of-moments estimators of α and β using the first two moments. Write your answer in terms of

$$m_1 = \frac{1}{n} \sum_{i=1}^n X_i \quad \text{and} \quad m_2 = \frac{1}{n} \sum_{i=1}^n X_i^2.$$

For $X \sim \text{Gamma}(\alpha, \beta)$,

$$\mathbb{E}[X] = \frac{\alpha}{\beta}, \quad \mathbb{E}[X^2] = \text{Var}(X) + \mathbb{E}[X]^2 = \frac{\alpha(\alpha+1)}{\beta^2}.$$

The moment equations are therefore

$$\frac{\alpha}{\beta} = m_1, \quad \frac{\alpha(\alpha+1)}{\beta^2} = m_2.$$

From the first equation, $\beta = \alpha/m_1$. Substitution into the second gives

$$m_2 = m_1^2 \frac{\alpha+1}{\alpha},$$

and hence

$$\hat{\alpha}_{\text{MOM}} = \frac{m_1^2}{m_2 - m_1^2}, \quad \hat{\beta}_{\text{MOM}} = \frac{m_1}{m_2 - m_1^2},$$

provided $m_2 - m_1^2 > 0$.

2. Let $X_1, \dots, X_n \stackrel{\text{i.i.d.}}{\sim} \mathcal{N}(\mu, \sigma^2)$.

- (a) Suppose σ^2 is known. Find the MLE of μ .
- (b) Suppose μ is known. Find the MLE of σ^2 .

- (a) Up to a term not depending on μ , the log-likelihood is

$$\ell(\mu) = -\frac{1}{2\sigma^2} \sum_{i=1}^n (x_i - \mu)^2.$$

Thus

$$\ell'(\mu) = \frac{1}{\sigma^2} \sum_{i=1}^n (x_i - \mu), \quad \ell''(\mu) = -\frac{n}{\sigma^2} < 0.$$

The unique maximizer is $\hat{\mu} = \bar{X}_n$.

- (b) Write $v = \sigma^2$ and $A = \sum_i (x_i - \mu)^2 > 0$. Up to an additive constant,

$$\ell(v) = -\frac{n}{2} \log v - \frac{A}{2v}.$$

Its derivative is

$$\ell'(v) = \frac{A - nv}{2v^2},$$

which is positive for $v < A/n$ and negative for $v > A/n$. Therefore,

$$\widehat{\sigma^2} = \frac{1}{n} \sum_{i=1}^n (X_i - \mu)^2.$$

3. In a simple model of inheritance, a person's genotype is **AA**, **aa**, or **Aa**. The two alleles are inherited independently and have the same distribution, with

$$\mathbb{P}_\theta(\mathbf{A}) = \theta, \quad \mathbb{P}_\theta(\mathbf{a}) = 1 - \theta,$$

where $0 < \theta < 1$. The order of the alleles does not matter, so **Aa** and **aA** represent the same genotype.

- (a) Compute $\mathbb{P}_\theta(\mathbf{Aa})$.
 (b) Encode the genotype as

$$X = \begin{cases} (1, 0, 0), & \text{if the genotype is AA,} \\ (0, 1, 0), & \text{if the genotype is aa,} \\ (0, 0, 1), & \text{if the genotype is Aa.} \end{cases}$$

Find the pmf $p_\theta(x)$ for $x \in \{(1, 0, 0), (0, 1, 0), (0, 0, 1)\}$.

- (c) A sample of 942 males from Ireland contains 501 of genotype **AA**, 83 of genotype **aa**, and 358 of genotype **Aa**. Find the MLE of θ .

- (a) There are two ordered ways to obtain the unordered genotype **Aa**, so

$$\mathbb{P}_\theta(\mathbf{Aa}) = 2\theta(1 - \theta).$$

- (b) For $x = (x_1, x_2, x_3)$ in the stated sample space,

$$p_\theta(x) = (\theta^2)^{x_1} ((1 - \theta)^2)^{x_2} (2\theta(1 - \theta))^{x_3}.$$

- (c) Up to a factor not depending on θ , the likelihood is

$$L(\theta) = (\theta^2)^{501} ((1 - \theta)^2)^{83} (2\theta(1 - \theta))^{358} \propto \theta^{1360} (1 - \theta)^{524}.$$

Thus

$$\ell'(\theta) = \frac{1360}{\theta} - \frac{524}{1 - \theta}.$$

Solving $\ell'(\theta) = 0$ gives

$$\widehat{\theta}_{\text{MLE}} = \frac{1360}{1360 + 524} = \frac{340}{471} \approx 0.72.$$

Since

$$\ell''(\theta) = -\frac{1360}{\theta^2} - \frac{524}{(1 - \theta)^2} < 0,$$

this critical point is the unique maximizer.

4. The lifetime X of a cell-phone battery has density

$$f_{\theta}(x) = K\theta^x \mathbb{1}_{\{x>0\}},$$

where $0 < \theta < 1$ is unknown and K is the normalizing constant. Let $X_1, \dots, X_n$ be independent observations from this model.

- (a) Show that $K = \log(1/\theta)$.
 (b) Rewrite the density in terms of $\eta = \log(1/\theta)$, and identify the resulting distribution of X (it is a named family). Write the statistical model in terms of η .
 (c) Find the MLE $\hat{\eta}$, then use $\theta = e^{-\eta}$ and invariance of the MLE to find $\hat{\theta}$.

- (a) Since

$$\int_0^{\infty} \theta^x dx = \int_0^{\infty} e^{-x \log(1/\theta)} dx = \frac{1}{\log(1/\theta)},$$

the normalizing constant is $K = \log(1/\theta)$.

- (b) With $\eta = \log(1/\theta)$, the density becomes

$$f_{\eta}(x) = \eta e^{-\eta x} \mathbb{1}_{\{x>0\}}.$$

Thus X has an exponential distribution with rate η . In terms of the new parameter, the statistical model for the sample is

$$\{\text{Exp}(\eta) : \eta > 0\}.$$

- (c) In terms of η , the likelihood is

$$L(\eta) = \eta^n e^{-\eta \sum_{i=1}^n X_i}.$$

The log-likelihood and its first two derivatives are

$$\ell(\eta) = n \log \eta - \eta \sum_{i=1}^n X_i, \quad \ell'(\eta) = \frac{n}{\eta} - \sum_{i=1}^n X_i, \quad \ell''(\eta) = -\frac{n}{\eta^2} < 0.$$

Hence $\hat{\eta} = 1/\bar{X}_n$, and invariance gives

$$\hat{\theta}_{\text{MLE}} = \exp\left(-\frac{1}{\bar{X}_n}\right).$$

5. Let $X_1, \dots, X_n \stackrel{\text{i.i.d.}}{\sim} \text{Unif}(\theta, \theta + 1)$, where $\theta \in \mathbb{R}$. Find the set of all maximum likelihood estimators of θ (there won't be just one MLE!).

The likelihood is

$$L(\theta \mid \underline{x}) = \mathbb{1}_{\{x_{(n)} - 1 \leq \theta \leq x_{(1)}\}}.$$

Its maximum value is one, and every

$$\theta \in [x_{(n)} - 1, x_{(1)}]$$

is a maximizer. Except when the interval collapses to a point, the MLE is not unique.

Evaluating estimators

6. Let $X_1, \dots, X_n \stackrel{\text{i.i.d.}}{\sim} \text{Unif}(0, \theta)$, where $\theta > 0$. Consider the method-of-moments estimator and the maximum likelihood estimator

$$\hat{\theta}_{\text{MoM}} = 2\bar{X}_n, \quad \hat{\theta}_{\text{MLE}} = X_{(n)} = \max(X_1, \dots, X_n).$$

- (a) Compute the bias, variance, and MSE of $\hat{\theta}_{\text{MoM}}$.
- (b) Compute the cdf and then the pdf of $\hat{\theta}_{\text{MLE}}$.
- (c) Compute the bias, variance, and MSE of $\hat{\theta}_{\text{MLE}}$.
- (d) Determine for which n the MSE of $\hat{\theta}_{\text{MLE}}$ is smaller than that of $\hat{\theta}_{\text{MoM}}$ for every $\theta > 0$.

- (a) Since $\mathbb{E}_\theta[X_i] = \theta/2$ and $\text{Var}_\theta(X_i) = \theta^2/12$,

$$\mathbb{E}_\theta[2\bar{X}_n] = \theta, \quad \text{Var}_\theta(2\bar{X}_n) = 4 \frac{\theta^2}{12n} = \frac{\theta^2}{3n}.$$

Thus $\hat{\theta}_{\text{MoM}}$ is unbiased and

$$\text{MSE}_\theta(\hat{\theta}_{\text{MoM}}) = \frac{\theta^2}{3n}.$$

- (b) Let $M = X_{(n)} = \hat{\theta}_{\text{MLE}}$. For $0 \leq m \leq \theta$, independence gives

$$\mathbb{P}_\theta(M \leq m) = \mathbb{P}_\theta(X_1 \leq m, \dots, X_n \leq m) = \left(\frac{m}{\theta}\right)^n.$$

Therefore,

$$F_{M,\theta}(m) = \begin{cases} 0, & m < 0, \\ (m/\theta)^n, & 0 \leq m \leq \theta, \\ 1, & m > \theta. \end{cases}$$

Differentiating on the support gives

$$f_{M,\theta}(m) = \frac{nm^{n-1}}{\theta^n} \mathbb{1}_{\{0 < m < \theta\}}.$$

- (c) Using this density, for $k = 1, 2$,

$$\mathbb{E}_\theta[M^k] = \int_0^\theta m^k \frac{nm^{n-1}}{\theta^n} dm = \frac{n}{n+k} \theta^k.$$

It follows that

$$\begin{aligned} \text{Bias}_\theta(M) &= \frac{n}{n+1} \theta - \theta = -\frac{\theta}{n+1}, \\ \text{Var}_\theta(M) &= \frac{n}{n+2} \theta^2 - \left(\frac{n}{n+1} \theta\right)^2 = \frac{n\theta^2}{(n+1)^2(n+2)}. \end{aligned}$$

The sample maximum is biased downward, which makes sense because it lies below θ with probability one. Its MSE is

$$\text{MSE}_\theta(M) = \frac{n\theta^2}{(n+1)^2(n+2)} + \frac{\theta^2}{(n+1)^2} = \frac{2\theta^2}{(n+1)(n+2)}.$$

(d) Comparing the two MSEs,

$$\begin{aligned}\text{MSE}_\theta(\hat{\theta}_{\text{MLE}}) < \text{MSE}_\theta(\hat{\theta}_{\text{MoM}}) &\iff \frac{2}{(n+1)(n+2)} < \frac{1}{3n} \\ &\iff (n-1)(n-2) > 0.\end{aligned}$$

Thus, for $n \geq 3$, the biased MLE has smaller MSE than the unbiased method-of-moments estimator for every $\theta > 0$. For $n = 1$ or $n = 2$, their MSEs are equal.

7. Let $X_1, \dots, X_n$ be i.i.d. random variables with density

$$f_\lambda(x) = \frac{1}{\lambda} e^{-x/\lambda} \mathbb{1}_{\{x \geq 0\}},$$

where $\lambda > 0$ is unknown.

- (a) Find the distribution of $X_{(1)} = \min_i X_i$ (Hint: use the trick from HW1, Problem 10).
- (b) Use the previous part to give an unbiased estimator of λ .
- (c) Compute the MSE of the estimator from part (b).
- (d) Compute the MSE of $\bar{X}_n$. Which of the two estimators has smaller MSE when $n > 1$?

(a) For $x \geq 0$,

$$\mathbb{P}_\lambda(X_{(1)} \geq x) = \mathbb{P}_\lambda(X_1 \geq x, \dots, X_n \geq x) = \mathbb{P}_\lambda(X_1 \geq x)^n = e^{-nx/\lambda}.$$

Therefore $X_{(1)}$ has an exponential distribution with mean λ/n .

- (b) Since $\mathbb{E}_\lambda[X_{(1)}] = \lambda/n$, the estimator $nX_{(1)}$ is unbiased for λ .
- (c) Because the estimator is unbiased,

$$\text{MSE}_\lambda(nX_{(1)}) = \text{Var}_\lambda(nX_{(1)}) = n^2 \left(\frac{\lambda}{n} \right)^2 = \lambda^2.$$

(d) Since $\bar{X}_n$ is unbiased for λ ,

$$\text{MSE}_\lambda(\bar{X}_n) = \text{Var}_\lambda(\bar{X}_n) = \frac{\lambda^2}{n}.$$

Thus $\bar{X}_n$ has smaller MSE when $n > 1$.